

Can Model-in-the-Loop Annotation Be Safely Scaled? Auditing Suggestion-Conditioned Label Noise in Human–AI Data Pipelines

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Abstract. As machine learning increasingly depends on labeled data, model-in-the-loop annotation has become the standard workflow for producing perception data in autonomous driving. But, when AI suggestions are wrong, annotators often accept them due to anchoring and automation bias, turning errors into human-confirmed labels regardless of confidence. We name this type of noise Suggestion-Conditioned Label Noise (SCLN).

To study this problem, we first build AILA-Bench, a benchmark for 2D object detection: we recruit 370 annotators to label 100,000 images from BDD100K under three conditions, Human-only, AI-assisted, and AI-assisted+confidence, and we record fine-grained interaction traces and AI error types. To address this problem, we formalize the counterfactual question of whether an annotator would still make the same error had they not seen the AI suggestion as a risk score SCLNScore, and we give the assumptions under which this score is identifiable. We then combine the annotations with ground truth to train two probabilistic models, p_1 and p_0 , with XGBoost to estimate this difference.

Experiments show that AI assistance introduces a different error structure and only gives a limited reduction in overall error rate. Compared with Human-only annotation, whose FDER is 31.5%, AI-assisted annotation with confidence decreases FDER to 30.8%. The main change is in the distribution of errors: localization errors rise from 3.9% in Human-only annotation to 13.9% under AI-assisted annotation. Among 567,963 source-linked final labels, 95,601 are linked to wrong AI suggestions, and 72.5% inherit the same error type. SCLNScore detects these AI-induced errors more effectively than confidence-based review, reaching AUPRC 0.472 and a $2.322\times$ gain at a 1% review budget; after transfer to nuImages, it remains above random with AUPRC 0.418 versus a 0.308 baseline. Demo: aila-bench.github.io/AILA-Bench.

Keywords: label noise · automation bias · human-AI annotation · model-in-the-loop · object detection · data quality auditing · counterfactual inference · autonomous driving

1 Introduction

The upper performance bound of autonomous-driving perception models largely depends on the label quality of their training data. High-quality detection annotation is costly, so industry widely uses a model-in-the-loop annotation pipeline [34]: an existing detector first generates candidate bounding boxes and classes, and annotators then edit labels one by one. This pipeline is attractive because it can reduce manual box drawing and scale annotation. More importantly, it is often treated as keeping a human quality-control step, because each label is confirmed by a person, and the final data are therefore assumed to be reliable.

However, this expectation is not consistent with long-standing findings in cognitive science and human factors. When people face suggestions from automated systems, they often show anchoring effects [32] and automation bias [28,19]: they may over-trust system outputs, reduce independent checking, and make errors when the system is wrong. Studies on human-AI decision making further show that over-reliance on artificial intelligence (AI) suggestions is hard to remove even when explanations or confidence scores are provided. Explanations may instead increase the probability that people accept suggestions, regardless of whether the suggestions are correct [1,3,33]. In annotation settings, non-expert annotators accept about 68% to 73% of AI suggestions, and the acceptance rate is almost unrelated to suggestion quality [2]. Crowd annotators also adopt many labels generated by large models, causing the label distribution to shift noticeably from the no-suggestion baseline [27].

When these mechanisms are placed back into the annotation pipeline, the consequence is direct: when AI pre-annotations are wrong, the human confirmation step may fail regardless of confidence level, and annotators may confirm AI errors as final labels. The resulting noise is not random human slips, but structured noise systematically induced by AI suggestions. We call this mechanism Suggestion-Conditioned Label Noise (SCLN), and its instance in two-dimensional (2D) detection is called SCLN-Det. Existing noise models do not cover SCLN. Classical work models noise as class-dependent transitions or instance-dependent noise that depends only on the sample itself [36]. Even studies that emphasize the difference between real human noise and synthetic noise still focus on noise produced by independent human annotation [35]. In contrast, SCLN depends jointly on the ground-truth sample, the AI suggestion, and how the suggestion is shown. The same image and the same object can therefore have systematically different error distributions with and without AI suggestions. To our knowledge, existing work on label noise and data auditing, such as Confident Learning [17] and detection-oriented label correction [21], has not modeled and audited this human-AI interaction noise as a separate object.

Studying SCLN first requires a dataset that can make it observable. To this end, we build the AI-in-the-Loop Annotation Benchmark (AILA-Bench): 370 annotators label 100,000 images from BDD100K under three conditions, Human-only, AI-assisted, and AI-assisted with confidence. We record fine-grained interaction traces, including action type, review time, and box edit distance, as well as AI error types. The three-condition comparison allows us to directly compare the

error distributions for the same object with and without AI suggestions, making the question of whether AI leads humans astray observable and reproducible.

On top of this benchmark, the main difficulty in auditing SCLN is attribution. An incorrect label confirmed by a human may come from the intrinsic difficulty of the task, or it may come from the influence of the AI suggestion. The ideal criterion for separating the two is a counterfactual question: would the annotator have made the same error if they had not seen the AI suggestion? Since we cannot observe both outcomes for the same annotation event, we define SCLNScore in Section 3.1 as a risk difference between the observed human-AI condition and a Human-only counterfactual baseline. Labels with larger scores are more likely to reflect AI-induced noise and are therefore prioritized for re-review.

We conduct experiments on AILA-Bench around four research questions (RQs). RQ1 asks whether errors in AI-assisted annotation differ from those in Human-only annotation. RQ2 asks whether final labels inherit wrong AI suggestions. RQ3 asks whether SCLNScore outperforms traditional re-review strategies. RQ4 asks whether SCLNScore generalizes across datasets. The current BDD100K export shows that AI assistance changes the error structure rather than simply improving label quality. Human-only has an FDER of 31.5%, AI-assisted has 32.9%, and AI-assisted annotation with confidence display has 30.8%, so the overall error-rate reduction from confidence display is limited. The sharper effect appears in error composition: BBox error rises from 3.9% in Human-only annotation to 13.9% under AI-assisted annotation and remains 11.5% even when confidence is shown. Among 567,963 source-linked final labels, 95,601 are linked to wrong AI suggestions, and 72.5% inherit the same error type. Confidence-based re-review is close to random, whereas SCLNScore reaches AUROC 0.665, AUPRC 0.472, Precision@1% of 0.738, and a 2.322 cost-normalized gain under a 1% review budget. A nuImages transfer check with 9,752 images reaches AUROC 0.607 and AUPRC 0.418, lower than BDD100K but still above the random AUPRC baseline of 0.308. Detailed results are reported in Section 4.

The contributions of this paper are summarized as follows.

- We are the first to identify and formalize SCLN in model-in-the-loop annotation, and we distinguish it from pure human noise and instance-dependent noise.
- We introduce AILA-Bench, the first systematic benchmark for SCLN. With 370 annotators labeling 100,000 BDD100K images under Human-only, AI-assisted, and AI-assisted with confidence display, AILA-Bench shows that AI assistance introduces a different error distribution and only limited error-rate reduction.
- We propose SCLNScore, the first auditing method for prioritizing SCLN under limited re-review budgets. It reaches AUPRC 0.472 and a 2.322 cost-normalized gain at a 1% review budget on BDD100K, and remains above random after transfer to nuImages with AUPRC 0.418.

2 Related Work

2.1 Label Noise

Label noise is commonly studied as either synthetic or real-world noise. Synthetic noise corrupts clean labels under a predefined process, such as uniform flipping, class-conditional flipping, or instance-dependent flipping [20,29]. Real-world noise arises from the annotation process itself. Wei et al. [35] re-collected human labels for CIFAR and showed that real human noise depends on the specific image and differs from synthetic noise. Instance-dependent noise models also make the noise rate depend on the sample [36]. However, to our knowledge, existing work on label noise and data auditing has not modeled or audited this form of human-AI interaction noise induced by AI suggestions.

Existing methods for learning with or detecting label noise fall into three main groups. Loss-correction methods estimate a noise transition matrix and adjust the training loss accordingly [20]. Sample-selection methods retain examples that are more likely to be clean, for example by selecting small-loss samples [10] or by separating clean and noisy data through the loss distribution [13]; related methods update models only on examples where two networks disagree [16,38]. Label-error detection methods directly identify mislabeled examples [17,18].

The AQUA benchmark compares such detectors across multiple data types under realistic noise [8]. For object detection, Ryoo et al. [25] group detection noise into wrong classes, wrong locations, missing objects, and spurious boxes. Other work trains detectors that tolerate noisy or loose boxes [5,14], studies spatial noise in segmentation masks [9], or detects and repairs label errors in detection datasets [21,31]. These methods can identify which labels are wrong, but they do not explain where the errors come from. SCLNScore further traces each error to its source, separating AI-induced noise from annotator mistakes.

2.2 AI-Assisted Human Annotation

Model pre-labeling followed by human review has become common in large-scale annotation pipelines [34,22,7,11]. Pre-annotation models mainly come from three detector families. Single-stage detectors such as YOLO predict boxes and classes in one pass, making them fast enough for interactive labeling [23,12]. Two-stage detectors such as Faster R-CNN first propose regions and then refine them, trading speed for accuracy [24]. Open-vocabulary detectors such as Grounding DINO use free-form text prompts to detect new classes without retraining [15].

In visual perception tasks, these suggestions are usually bounding boxes, and their errors can directly propagate into detectors trained on the reviewed labels [26]. The core problem is that reviewers often trust the machine too much. Anchoring makes people rely on the first value they see [32], while automation bias makes them miss errors and accept incorrect suggestions, even when they are trained experts [28,19].

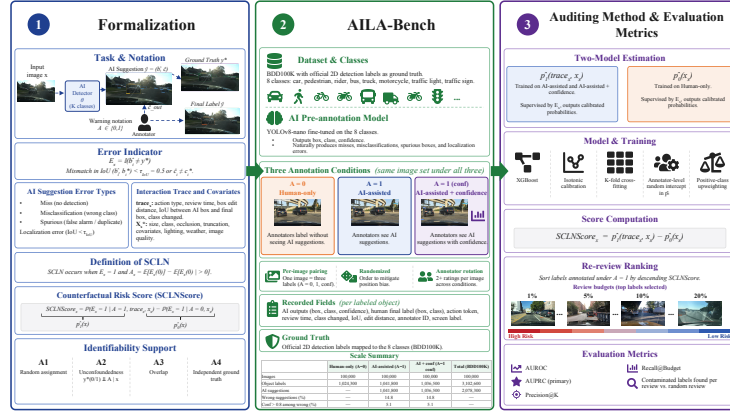


Fig. 1. Workflow of the proposed method.

3 Method

This section follows the workflow in Figure 1 and has three parts. §3.1 formalizes SCLN as a counterfactual risk score SCLNScore. §3.2 builds AILA-Bench, which provides the annotations, interaction traces, and ground truth needed to estimate that score. §3.3 gives the estimation procedure, the re-review ranking, and the evaluation metrics.

3.1 Formalization

Task and notation. The basic unit of annotation is an object label $y = (\mathbf{b}, c)$, where $\mathbf{b} \in \mathbb{R}^4$ is a bounding box and $c \in \mathcal{C}$ is one of 8 safety-relevant classes detailed in §3.2. In a model-in-the-loop pipeline, a detector first outputs a suggestion $\hat{y} = (\hat{\mathbf{b}}, \hat{c})$ together with a confidence score $\hat{c}_{\text{conf}} \in [0, 1]$. An annotator then produces a final label $\tilde{y} = (\tilde{\mathbf{b}}, \tilde{c})$ under a viewing condition $A \in \{0, 1\}$, where $A=1$ means the AI suggestion was shown and $A=0$ means it was withheld. The ground truth $y^* = (\mathbf{b}^*, c^*)$ is taken from the official annotations of each dataset, see §3.2. For object i , the error indicator of the final label is

$$E_i = \mathbb{I}[\tilde{y}_i \neq y_i^*], \quad \tilde{y}_i \neq y_i^* \iff \text{IoU}(\tilde{\mathbf{b}}_i, \mathbf{b}_i^*) < \tau_{\text{IoU}} \text{ or } \tilde{c}_i \neq c_i^*, \quad (1)$$

with $\tau_{\text{IoU}} = 0.5$. We use four primary error types, matching the error composition used in RQ1. A *BBox* error means the predicted or final box is assigned to the right object class but its localization falls below the IoU threshold. A *Class* error means the box is matched to a real object but the class is wrong. A *false positive (FP)* means a predicted or final box does not match any real object, including duplicate boxes for the same target. A *Miss* means a real object has no matched box. We denote the interaction trace recorded under the AI condition by trace_i , which includes the action type, review time, box edit distance, IoU between

the AI box and the final box, and whether the class was changed. We denote the intrinsic covariates of the sample and scene by x_i , which include object size, class, occlusion, truncation, crowding, lighting, weather, and image quality. Note that x_i is available under any condition, while trace_i exists only when the AI suggestion was shown.

Definition of SCLN. A final label $\tilde{y}_i$ confirmed by a human contains SCLN if two conditions hold. First, $E_i=1$, i.e., the label is wrong relative to the ground truth. Second, the error is attributable to the AI suggestion, meaning that in the counterfactual Human-only setting the same object would be noticeably less likely to be mislabeled. Let $E_i(1)$ and $E_i(0)$ denote the potential outcomes with and without the AI suggestion. The suggestion-induced effect for a single object is $\Delta_i = \mathbb{E}[E_i(1)] - \mathbb{E}[E_i(0)]$. $\Delta_i > 0$ means the AI suggestion raised the error probability for that object, i.e., it led the annotator astray. $\Delta_i < 0$ means the suggestion helped, for instance by pointing out a missed object. SCLN concerns labels where Δ_i is clearly positive. Classical noise models write the transition probability as $P(\tilde{c} \mid c^*)$ or $P(\tilde{y} \mid x)$, corresponding to class-dependent and instance-dependent noise. The transition probability of SCLN is $P(\tilde{y}_i \mid y_i^*, x_i, \hat{y}_i, \hat{c}_{\text{conf},i}, A)$, which depends explicitly on the AI suggestion and the viewing condition. Holding the ground truth and sample fixed while changing A changes the error distribution. This dependence structure is absent from the two classical noise types and is what our three-condition design is built to measure. **The counterfactual risk score SCLNScore.** The individual effect Δ_i requires observing both potential outcomes for the same annotation event, which is impossible. We instead estimate a rankable risk score for auditing:

$$\boxed{\text{SCLNScore}_i = \underbrace{P(E_i=1 \mid A=1, \text{trace}_i, x_i)}_{p_1(i)} - \underbrace{P(E_i=1 \mid A=0, x_i)}_{p_0(i)}} \quad (2)$$

The two terms condition on different information, and this asymmetry is deliberate. $p_1(i)$ is the posterior error risk under the observed human–AI interaction. It draws on post-treatment interaction traces: a fast confirm with almost no edit to the AI box is a strong signal of contamination. $p_0(i)$ is the baseline error risk under independent annotation, using only pre-treatment covariates that capture how hard the sample is. The difference between them measures the extra error risk brought by the human–AI interaction beyond what the task difficulty alone can explain. A higher score makes AI-induced contamination more likely, so labels should be re-reviewed in descending order of SCLNScore.

Identifiability and design support. p_1 is trained directly on AI-assisted labels with ground-truth supervision, while p_0 transports Human-only error risk to samples annotated under AI assistance. This transport is supported by the AILA-Bench design: image–annotator pairs are randomly assigned to conditions, the same image set is covered across conditions, Human-only data provide overlap over object and scene covariates, and the ground truth is independent of the audited annotations. Under these conditions, SCLNScore is used as a rank-consistent proxy for AI-induced error risk, rather than a per-sample causal effect.

3.2 AILA-Bench

AILA-Bench makes SCLN measurable. At its core is a three-condition annotation design with per-image pairing and fine-grained interaction traces.

Dataset and classes. We use the full BDD100K detection set [37]. Its diversity in geography, weather, and time of day lets us study how scene difficulty affects SCLN. BDD100K ships with official 2D detection labels, which serve directly as the ground truth y^* . Cross-dataset tests use nuImages [4] and the Waymo Open Dataset 2D subset [30], described in Section 4. We use eight safety-relevant classes: car, pedestrian, rider, bus, truck, motorcycle, traffic light, and traffic sign. These classes cover frequent objects, long-tail categories, small objects, and safety-critical road agents. Class frequency and object size serve as covariates for both p_0 and p_1 and as stratification dimensions.

AI pre-annotation model. We instantiate the pre-annotation step with YOLOv8 nano, a lightweight single-stage detector from the YOLOv8 family [12], fine-tuned on the eight study classes. The detector outputs a candidate box, class label, and confidence score ($\hat{\mathbf{b}}, \hat{c}, \hat{c}_{\text{conf}}$) for each suggestion. We use this setting because lightweight detectors are commonly used in interactive annotation workflows, where latency and deployment cost matter. After matching suggestions and final labels to the ground truth, we categorize suggestion errors into BBox, Class, FP, and Miss using the taxonomy in §3.1. For RQ2, we group source suggestions by confidence bin and report the observed error frequency and inheritance pattern within each bin.

Three annotation conditions. The same set of images is labeled under three conditions. Human-only shows the raw image with no AI boxes and serves as the independent-annotation baseline for training p_0 . AI-assisted overlays the AI boxes and class labels on the image, to see whether suggestions shift error patterns. AI-assisted+confidence shows the confidence score, to see whether displaying confidence further increases error acceptance. Four practices keep the conditions comparable. First, per-image pairing: each image is labeled under all three conditions, so the same object’s error can be compared across conditions. This supports RQ1, RQ2, and the transport of p_0 . Second, no annotator sees the same image in more than one version; we rotate annotators across conditions with a Latin square to balance individual skill. Third, the assignment of images and annotators to conditions is randomized, which underpins Assumption A1. Fourth, annotators are not told the purpose of the study and all receive the same training and qualification test, with annotator IDs recorded as random effects.

Recorded fields and AI error types. For each object we record both the AI output and the human operation, which together form the interaction trace trace_i . From the AI side we record the predicted box, class, and confidence. From the human side we record the final box and class, the action taken, the review time, whether the class was changed, and the IoU and edit distance between the AI box and the final box. Every object also carries an annotator ID and a scene label for stratified analysis. The AI suggestion errors follow the four-type taxonomy of BBox, Class, FP, and Miss defined in §3.1, and are used for the error-pattern analysis and contamination attribution in RQ1.

Table 1. AILA-Bench scale by annotation condition.

	Human-only	AI-assisted	AI+conf	Total
Images	100,000	100,000	100,000	100,000
Object labels	1,024,300	1,041,800	1,036,500	3,102,600
AI suggestions	—	1,041,800	1,036,500	2,078,300
Annotators	123	123	124	370

Ground truth. The ground truth y^* is based on the official 2D detection annotations shipped with each dataset. To assess the reliability of the reference labels, three experts independently reviewed a random sample of 2,000 official BDD100K annotations, with disagreements resolved by majority vote. The estimated error rate of the official reference labels is 4.5%. The only adaptation we make is to map the class labels to our 8 classes.

Scale summary. Table 1 reports the size of AILA-Bench and the distribution across the three conditions.

3.3 Auditing Method and Evaluation Metrics

We estimate the difference in Equation (2) with two probabilistic models:

$$\widehat{\text{SCLNScore}}_i = \hat{p}_1(\text{trace}_i, x_i) - \hat{p}_0(x_i).$$

$\hat{p}_1$ is trained on the AI-assisted and AI-assisted+confidence data, and $\hat{p}_0$ is trained on the Human-only data. Both are supervised by the ground-truth error E_i and both output calibrated probabilities. At audit time, we compute $\hat{p}_1$ from the real interaction trace, $\hat{p}_0$ from the sample covariates, subtract, and sort in descending order. This divide-and-conquer structure matches the T-learner in causal inference: p_1 uses post-treatment interaction features while p_0 uses only pre-treatment covariates, so the two models separate the risk contributed by the interaction from the risk contributed by task difficulty.

Model and training. Both p_1 and p_0 are trained with XGBoost [6], which handles the mix of numeric and categorical features in our setting and captures nonlinear relationships among them. After training we apply isotonic regression to calibrate both models so that their outputs sit on the same scale and the subtraction is meaningful. Because E_i comes from the ground truth and p_1 uses interaction features that correlate with E_i , training and scoring on the full dataset would let the model memorize the labels it has seen. We avoid this with K -fold cross-fitting: for each fold, both models are trained on the remaining folds and produce out-of-fold predictions. We also add an annotator-level random intercept to p_1 to absorb individual skill differences, so that a generally less accurate annotator is not mistaken for someone being led astray by the AI. During training we upweight the positive class to handle the sparsity of contaminated labels. At re-review time, given a budget, say 1%, 5%, 10%, or 20% of all labels, we take the top-scoring batch and send it to a human reviewer. The reviewer’s judgment can be fed back into the training data so that p_1 improves over

time. The subsequent experiments treat the AI-contaminated labels defined in §3.1 as positives and measure audit quality with five metrics. AUROC captures overall ranking quality. AUPRC, more sensitive when positives are sparse, is the primary metric. Precision@ K is the fraction of true positives among the top K scored labels. Recall@Budget is the fraction of all contaminated labels recovered under a given review budget. The last metric is the number of contaminated labels found per review, reported as a multiple of random review, which directly answers the question of whether each extra label reviewed is worth the cost.

4 Evaluation

4.1 Experimental Setup and Evaluation Metrics

We organize the experiments around the four research questions (RQs) defined above. RQ1 tests whether errors under the AI-assisted condition differ from those under the Human-only condition. RQ2 tests whether final human labels are more likely to follow and inherit an AI suggestion when the suggestion is wrong. RQ3 tests whether SCLNScore outperforms random, confidence-based, and disagreement-based review policies. RQ4 tests whether SCLNScore transfers from BDD100K to other datasets. All error judgments use the official two-dimensional detection annotations as gold labels, with an Intersection over Union (IoU) threshold of 0.5.

We use the following metrics. Final Detection Error Rate (FDER) is the fraction of final labels that disagree with gold labels, and accuracy (Acc.) is defined as $1 - \text{FDER}$. Error composition decomposes FDER into BBox, Class, false positive (FP), Miss, and Other, where Other is the remaining error mass not covered by the listed types. In RQ2, AI conf. denotes the confidence bin of the source AI suggestion, and AI-linked labels are final labels that can be traced to a source AI suggestion. AI wrong is the fraction of source AI suggestions that are erroneous. Same error inherited is the fraction of erroneous AI suggestions for which the final label has the same error type. Final error is the error rate of final labels in the confidence bin. In RQ3, Area Under the Receiver Operating Characteristic Curve (AUROC) measures the ranking ability of SCLNScore for contaminated and non-contaminated labels. Area Under the Precision-Recall Curve (AUPRC) is the main ranking metric when positive labels are sparse. Precision@ K is the fraction of contaminated labels among the top K ranked labels, and Recall@Budget is the fraction of all positive labels recovered under a given review budget. Cost-normalized gain is the Precision@Budget of a policy divided by the precision of random review under the same budget, and values above 1 indicate a gain over random review. RQ4 uses the same ranking metrics to evaluate the transfer performance of BDD-trained SCLNScore on nuImages.

4.2 RQ1: Do Errors in AI-assisted Annotation Differ from Human-only Annotation?

This experiment compares FDER, accuracy, and error composition under the three interface conditions. The label counts in this section refer to evaluable

Table 2. Task scale, final detection error rate, and error composition on the current BDD100K pilot. Values are percentages except Tasks and Labels. Other is the remainder of FDER not covered by the displayed error types.

Condition	Tasks	Labels	Acc.	FDER	BBox	Class	FP	Miss	Other
Human-only	100,000	837,153	68.5	31.5	3.9	0.7	12.2	11.6	2.9
AI-assisted	100,000	903,455	67.1	32.9	13.9	3.8	10.1	0.3	4.8
AI+conf	100,000	903,455	69.2	30.8	11.5	4.1	10.0	0.2	5.0

final labels after class filtering and ground-truth matching, and therefore differ from the raw object-label scale reported in Table 1.

Table 2 shows that the error distribution of AI-assisted annotation clearly differs from Human-only annotation. The three conditions have similar FDER values: Human-only at 31.5%, AI-assisted at 32.9%, and AI-assisted+confidence at 30.8%. The error composition, however, is different. In Human-only, Miss is 11.6% and FP is 12.2%; together, they account for 75.6% of FDER. BBox error is only 3.9%, or 12.4% of FDER. This pattern shows that Human-only errors mainly come from missing objects and extra boxes.

Under AI-assisted conditions, the error mass moves toward the quality of existing boxes. Standard AI-assisted has only 0.3% Miss, but BBox error reaches 13.9%, or 42.2% of FDER. AI-assisted+confidence also has only 0.2% Miss, while BBox error is 11.5%, or 37.3% of FDER. Thus, the main difference between AI conditions and Human-only is not a small change in FDER, but a redistribution of error types: AI suggestions greatly reduce missed-label errors while making localization and class errors a larger share of final errors. After confidence is displayed, BBox error drops from 13.9% in standard AI-assisted to 11.5% in AI-assisted+confidence, but its error structure still differs from Human-only.

4.3 RQ2: Do Final Labels Inherit the Same Errors When AI Provides Wrong Suggestions?

We only count final labels that can be traced to a source AI suggestion, so this source-linked subset is smaller than the full evaluable-label set used in RQ1. If an AI suggestion is not correct and the final label has the same error type, we count it as same error inherited. This definition is more conservative than full causal attribution because it only covers final boxes with source links.

Table 3 shows that many wrong AI suggestions enter the final labels. Among 567,963 source-linked final labels, 95,601 come from wrong AI suggestions; among them, 69,278 keep the same error type as the source suggestion, giving an inheritance rate of 72.5%.

Confidence bins further show the conditional pattern of error inheritance. The low-confidence bin has the highest AI wrong rate, at 43.3%, which means low-confidence suggestions are more likely to be wrong. However, among suggestions that are already wrong, same error inherited rises from 66.4% in the

Table 3. Source-linked final labels grouped by AI confidence. SEI denotes same error inherited, meaning that the final label has the same error type as an erroneous AI suggestion.

AI conf.	AI-linked labels	AI wrong	SEI	Final error
0.25-0.50	97,097	43.3	66.4	38.9
0.50-0.75	197,318	18.9	72.6	21.8
0.75-1.00	273,548	6.0	87.5	7.8

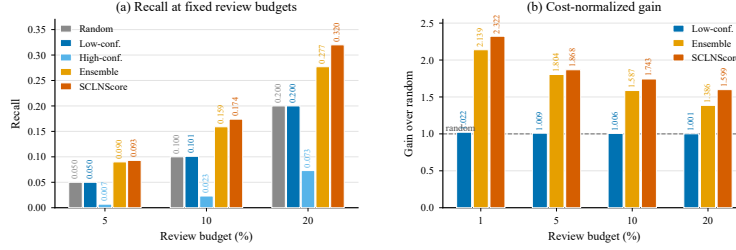


Fig. 2. Error retrieval under limited review budgets. The left panel shows how many contaminated labels each budget recovers, and the right panel shows the gain per unit budget relative to random review.

low-confidence bin to 87.5% in the high-confidence bin. High-confidence suggestions are less often wrong, but when they are wrong, the same error type is more likely to remain in the final label.

4.4 RQ3: Does SCLNScore Outperform Traditional Review Policies?

This experiment treats review as an error retrieval problem under a fixed budget: given a budget that only allows review of the top $b\%$ labels, a better ranking policy places more contaminated labels near the top, leading to higher Recall@Budget and cost-normalized gain. We compare Random, Low-confidence, High-confidence, Ensemble disagreement, and SCLNScore.

Figure 2 shows that SCLNScore mainly improves concentrated error detection under limited budgets. At 5%, 10%, and 20% budgets, random review recovers only about 5.0%, 10.0%, and 20.0% of contaminated labels. Under the same budgets, SCLNScore reaches Recall values of 9.3%, 17.4%, and 32.0%, and is consistently higher than random, confidence-based ranking, and ensemble disagreement. Low-confidence nearly overlaps with random, which shows that low confidence alone does not rank final errors effectively. By cost-normalized gain, SCLNScore reaches 2.322 times random review at a 1% budget and remains at 1.868, 1.743, and 1.599 times random review under 5%, 10%, and 20% budgets. Ensemble disagreement also improves review efficiency, but it is lower than SCLNScore under every budget.

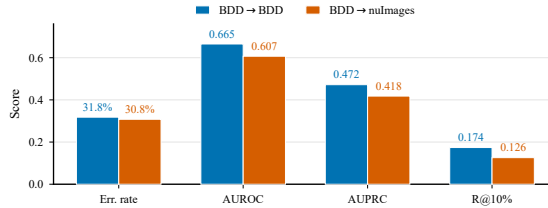


Fig. 3. Cross-dataset ranking results for BDD-trained SCLNScore. Err. rate also gives the AUPRC baseline for random ranking; higher AUROC, AUPRC, and R@10% indicate more effective error ranking.

4.5 RQ4: Can SCLNScore Transfer Across Datasets?

We directly apply the SCLNScore model trained on BDD100K to nuImages v1.0-test. The nuImages experiment uses official two-dimensional detection annotations as gold labels and YOLOv8 to generate suggestions, covering 9,752 images. We use the same export, matching, error labeling, and ranking pipeline as BDD100K, without retraining the scorer on nuImages.

RQ4 tests whether a BDD-trained scorer still provides a useful review ranking signal on a new dataset. As shown in Figure 3, the ranking metrics decrease after transfer to nuImages relative to BDD100K: AUROC drops from 0.665 to 0.607, AUPRC drops from 0.472 to 0.418, and R@10% drops from 0.174 to 0.126. This drop shows that cross-dataset transfer reduces the ranking power of SCLNScore. At the same time, BDD-trained SCLNScore still keeps an effective error ranking signal on nuImages. The Err. rate on nuImages is 30.8%, which is the AUPRC baseline for random ranking; the transferred AUPRC is 0.418, 11.0 percentage points above that baseline. In other words, even without retraining on nuImages, SCLNScore ranks erroneous labels ahead of random review.

5 Discussion

Can SCLN-Det Generalize Beyond 2D Detection and Autonomous Driving? SCLN is not limited to 2D detection or to autonomous driving. It is a general problem in any model-in-the-loop pipeline. The setup needs only three things: a person confirms an AI suggestion, there is a contrast between an AI-assisted and a Human-only condition, and an interaction trace is recorded during review. None of these is specific to bounding boxes or driving scenes. The same noise arises in instance segmentation, 3D detection, medical imaging, remote sensing, and content moderation, wherever a human confirms AI pre-annotations. In many of these domains a confirmed AI error is far more costly than in ours, so auditing SCLN matters even more. The main finding holds throughout: a human confirmation step does not remove AI-induced noise on its own, and auditing that noise needs a counterfactual score, not the AI confidence

alone. SCLN-Det gives a general way to define and audit this noise, and applying it beyond autonomous driving data is left for future work.

Where Is Error Risk Concentrated? The feature analysis suggests that final-label error risk is concentrated in specific object, AI-suggestion, and interaction patterns rather than being uniformly spread across labels, Object features are the strongest group: using only category and box area reaches AUPRC 0.566, and removing Object from the full model reduces AUPRC by 0.112. AI suggestion features are the next strongest signal, with group-only AUPRC 0.522 and a removal drop of 0.083. Trace and Interface+Time are still useful, reaching AUPRC 0.431 and 0.397, with removal drops of 0.031 and 0.018. At the single-feature level, category reaches AUPRC 0.507, source_ai_confidence reaches 0.463, event_count reaches 0.397, and review_time_ms reaches 0.384. The interaction traces also differ by interface: Human-only labels are mainly tied to box_create events, while AI-assisted settings show more box_edit, box_delete, confirm, and class_change events.

6 Conclusion

In this paper, we identify and formalize Suggestion-Conditioned Label Noise (SCLN). We build two things. AILA-Bench is a benchmark that makes SCLN observable under a three-condition design. SCLNScore is a counterfactual risk score that ranks labels by how likely an AI suggestion led the annotator astray. Our experiments show two findings. First, annotators inherit the same wrong AI error 72.5% of the time. Second, at a 1% review budget, SCLNScore finds about 2.3 times as many contaminated labels as random review, and it still transfers to an unseen dataset.

References

1. Bansal, G., Wu, T., Zhou, J., Fok, R., Nushi, B., Kamar, E., Ribeiro, M.T., Weld, D.: Does the whole exceed its parts? the effect of AI explanations on complementary team performance. In: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. pp. 1–16 (2021)
2. Beck, T., Lee, J.U., Viehmann, C., Maurer, M., Quiring, O., Gurevych, I.: Investigating label suggestions for opinion mining in german covid-19 social media. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers). pp. 1–13 (2021)
3. Bućinca, Z., Malaya, M.B., Gajos, K.Z.: To trust or to think: cognitive forcing functions can reduce overreliance on ai in ai-assisted decision-making. Proceedings of the ACM on Human-computer Interaction **5**(CSCW1), 1–21 (2021)
4. Caesar, H., Bankiti, V., Lang, A.H., Vora, S., Liong, V.E., Xu, Q., Krishnan, A., Pan, Y., Baldan, G., Beijbom, O.: nuScenes: A multimodal dataset for autonomous driving. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 11621–11631 (2020)

5. Chadwick, S., Newman, P.: Training object detectors with noisy data. In: 2019 IEEE Intelligent Vehicles Symposium (IV). pp. 1319–1325. IEEE (2019)
6. Chen, T., Guestrin, C.: Xgboost: A scalable tree boosting system. In: Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining. pp. 785–794 (2016)
7. Gilardi, F., Alizadeh, M., Kubli, M.: ChatGPT outperforms crowd workers for text-annotation tasks. Proceedings of the National Academy of Sciences (PNAS) **120**(30), e2305016120 (2023)
8. Goswami, M., Sanil, V., Choudhry, A., Srinivasan, A., Udompanyawit, C., Dubrawski, A.: Aqua: A benchmarking tool for label quality assessment. Advances in Neural Information Processing Systems **36**, 79792–79807 (2023)
9. Grad, E., Kimhi, M., Halika, L., Baskin, C.: Benchmarking label noise in instance segmentation: Spatial noise matters. arXiv e-prints pp. arXiv-2406 (2024)
10. Han, B., Yao, Q., Yu, X., Niu, G., Xu, M., Hu, W., Tsang, I., Sugiyama, M.: Co-teaching: Robust training of deep neural networks with extremely noisy labels. Advances in neural information processing systems **31** (2018)
11. He, Z., Huang, C.Y., Ding, C.K.C., Rohatgi, S., Huang, T.H.K.: If in a crowd-sourced data annotation pipeline, a gpt-4. In: Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems. pp. 1–25 (2024)
12. Jocher, G., Chaurasia, A., Qiu, J.: Ultralytics YOLO (version 8.0) [computer software]. <https://github.com/ultralytics/ultralytics>, accessed 18 August 2026 (2023)
13. Li, J., Socher, R., Hoi, S.C.: Dividemix: Learning with noisy labels as semi-supervised learning. arXiv preprint arXiv:2002.07394 (2020)
14. Li, J., Xiong, C., Socher, R., Hoi, S.: Towards noise-resistant object detection with noisy annotations. arXiv preprint arXiv:2003.01285 (2020)
15. Liu, S., Zeng, Z., Ren, T., Li, F., Zhang, H., Yang, J., Jiang, Q., Li, C., Yang, J., Su, H., Zhu, J., Zhang, L.: Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In: European Conference on Computer Vision (ECCV). pp. 38–55 (2024)
16. Malach, E., Shalev-Shwartz, S.: Decoupling “when to update” from “how to update”. In: Advances in Neural Information Processing Systems. vol. 30 (2017)
17. Northcutt, C., Jiang, L., Chuang, I.: Confident learning: Estimating uncertainty in dataset labels. Journal of Artificial Intelligence Research **70**, 1373–1411 (2021)
18. Northcutt, C.G., Athalye, A., Mueller, J.: Pervasive label errors in test sets destabilize machine learning benchmarks. In: Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks (2021)
19. Parasuraman, R., Manzey, D.H.: Complacency and bias in human use of automation: An attentional integration. Human Factors **52**(3), 381–410 (2010)
20. Patrini, G., Rozza, A., Krishna Menon, A., Nock, R., Qu, L.: Making deep neural networks robust to label noise: A loss correction approach. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 1944–1952 (2017)
21. Penquitt, S., Klees, J., Cakaj, R., Kondermann, D., Rottmann, M., Schmarje, L.: From label error detection to correction: a modular framework and benchmark for object detection datasets. arXiv preprint arXiv:2508.06556 (2025)
22. Ravi, N., Gabeur, V., Hu, Y.T., Hu, R., Ryali, C., Ma, T., Khedr, H., Rädle, R., Rolland, C., Gustafson, L., et al.: Sam 2: Segment anything in images and videos. In: International Conference on Learning Representations. vol. 2025, pp. 28085–28128 (2025)

23. Redmon, J., Divvala, S., Girshick, R., Farhadi, A.: You only look once: Unified, real-time object detection. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 779–788 (2016)
24. Ren, S., He, K., Girshick, R., Sun, J.: Faster r-cnn: Towards real-time object detection with region proposal networks. *Advances in neural information processing systems* **28** (2015)
25. Ryoo, K., Jo, Y., Lee, S., Kim, M., Jo, A., Kim, S.H., Kim, S., Lee, S.: Universal noise annotation: Unveiling the impact of noisy annotation on object detection. *arXiv preprint arXiv:2312.13822* (2023)
26. Saeeda, H., Johansson, T., Mohamad, M., Knauss, E.: Data annotation quality problems in ai-enabled perception system development. *arXiv preprint arXiv:2511.16410* (2025)
27. Schroeder, H., Roy, D., Kabbara, J.: Just put a human in the loop? investigating llm-assisted annotation for subjective tasks. In: *Findings of the Association for Computational Linguistics: ACL 2025*. pp. 25771–25795 (2025)
28. Skitka, L.J., Mosier, K.L., Burdick, M.: Does automation bias decision-making? *International Journal of Human-Computer Studies* **51**(5), 991–1006 (1999)
29. Song, H., Kim, M., Park, D., Shin, Y., Lee, J.G.: Learning from noisy labels with deep neural networks: A survey. *IEEE transactions on neural networks and learning systems* **34**(11), 8135–8153 (2022)
30. Sun, P., Kretzschmar, H., Dotiwalla, X., Chouard, A., Patnaik, V., Tsui, P., Guo, J., Zhou, Y., Chai, Y., Caine, B., et al.: Scalability in perception for autonomous driving: Waymo open dataset. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 2446–2454 (2020)
31. Tkachenko, U., Thyagarajan, A., Mueller, J.: ObjectLab: Automated diagnosis of mislabeled images in object detection data. In: *ICML Workshop on Data-centric Machine Learning Research (DMLR)* (2023)
32. Tversky, A., Kahneman, D.: Judgment under uncertainty: Heuristics and biases. *Science* **185**(4157), 1124–1131 (1974). <https://doi.org/10.1126/science.185.4157.1124>
33. Vasconcelos, H., Jörke, M., Grunde-McLaughlin, M., Gerstenberg, T., Bernstein, M.S., Krishna, R.: Explanations can reduce overreliance on AI systems during decision-making. *Proceedings of the ACM on Human-Computer Interaction* **7**(CSCW1), 1–38 (2023)
34. Wang, Y., Stevens, D., Shah, P., Jiang, W., Liu, M., Chen, X., Kuo, R., Li, N., Gong, B., Lee, D., et al.: Model-in-the-loop (milo): Accelerating multimodal ai data annotation with llms. *arXiv preprint arXiv:2409.10702* (2024)
35. Wei, J., Zhu, Z., Cheng, H., Liu, T., Niu, G., Liu, Y.: Learning with noisy labels revisited: A study using real-world human annotations. *arXiv preprint arXiv:2110.12088* (2021)
36. Xia, X., Liu, T., Han, B., Wang, N., Gong, M., Liu, H., Niu, G., Tao, D., Sugiyama, M.: Part-dependent label noise: Towards instance-dependent label noise. In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 33, pp. 7597–7610 (2020)
37. Yu, F., Chen, H., Wang, X., Xian, W., Chen, Y., Liu, F., Madhavan, V., Darrell, T.: BDD100K: A diverse driving dataset for heterogeneous multitask learning. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 2636–2645 (2020)
38. Yu, X., Han, B., Yao, J., Niu, G., Tsang, I., Sugiyama, M.: How does disagreement help generalization against label corruption? In: *International conference on machine learning*. pp. 7164–7173. PMLR (2019)